

From Static to Dynamic: Parameter Manipulation as Adaptive DDL Scaffolding in Creative Writing with Generative AI

Abstract: Generative AI tools now pervasively integrate into university classrooms, yet the field lacks empirical understanding of how technical AI configurations translate into adaptive Data-driven Learning (DDL) scaffolding. This gap is critical in creative writing where excessive structure can suppress authorship and insufficient support can stall exploration. This study investigates how large language model (LLM) parameter configuration functions as a pedagogical design lever for DDL scaffolding in poetry writing classrooms. The work contributes by rendering parameter effects explicit and measurable, enabling instructors to design responsive AI systems that dynamically adapt scaffolding to learner needs, creative agency, and satisfaction.

Key Words: Data-driven Learning; AI parameter configuration; Creative poetry writing; Adaptive scaffolding; Human-AI collaboration.

1. Introduction

With the rise of generative AI, scholars pose a challenge to the language teaching profession: Generative AI, allowing learners to bypass inductive pattern discovery processes, represents the death of Data-driven learning (Crosthwaite and Baisa 2023). This paper argues that generative AI may represent an evolution rather than replacement of DDL principles. We examine this proposition through AI-assisted poetry writing in university creative writing classrooms, where the tension between DDL principles and generative AI becomes visible and pedagogically consequential.

Data-Driven Learning (DDL), put forward by Johns (1991) as the use of concordance data in language learning, has long valued inductive pattern discovery over deductive rule presentation. By engaging learners directly with corpus evidence,

DDL shifts pedagogy from teacher-transmitted knowledge to learner-discovered insights (Boulton 2012). The power of DDL lies in learner autonomy and evidence-based discovery, aligning with constructivist and sociocultural perspectives on learning (Lantolf and Thorne 2006).

To answer whether AI-assisted writing constitutes DDL, we must challenge narrow conceptions of DDL as merely keyword matching or concordance consultation (Sinclair 1991). At its core, DDL is data-driven decision making in language learning: learners form hypotheses about language use, test them against patterns in data, and refine their output. Traditional DDL accomplishes this through explicit corpus queries and concordance searches (Cobb 2013). LLMs are trained on massive corpora and generate outputs by sampling from learned distributions of language patterns (Vaswani et al. 2017; Radford et al. 2019). When AI generates a poetic line or suggests an alternative phrasing, it functions as a data-driven pattern provider, even though the mechanism differs from explicit corpus search. In short, AI can serve as a new DDL interface if learners treat its outputs as linguistic data to notice, analyze, and revise.

Why poetry, and why AI-assisted poetry, count as DDL requires explicit justification. Poetry writing in university classrooms is not ineffable but effable. Heidegger (1984) frames poetry as the opposite of machine logic, defining poetry as one of the purest form of human artistic expression and creativity, something vastly different from machinery operation which based on scientific approaches, logic, and reasoning. At the very core of this type of thinking is a deep-rooted idea that science and artistic expression are essentially incompatible. However, in terms of writing process, poetry composition could be viewed as a certain type of reasoning and problem-solving, while, according to Strehovec (2023), artistic creation “can be understood as a cognitive and goal-oriented activity.” Strehovec’s study views poets as the role of a researcher and his/her writing process as a sophisticated labor of verifying scientific approaches to individual components of reality. It is, much like operating a program, a hard work that involves a series of rational, even methodology-supported procedures.

Our stance is that machine influence is pedagogically valid when it is bounded and transparent: AI can provide pattern data and alternative options, while human writers remain responsible for selection, rejection, and revision. This is especially important for beginners in writing, who often have ideas and affective goals but lack linguistic resources to realize them. For these writers, AI-assisted poetry can still be DDL when the process foregrounds noticing, testing, and reflective decision making rather than substitution or ghostwriting.

In collaborative creative writing, the procedure combines student decisions (theme, emotion, style, form) with AI pattern-based text generation. Students exercise agency in conceptual and creative dimensions, while AI provides data-driven linguistic scaffolding, paralleling traditional DDL where learners formulate queries and receive evidence from corpus data. Therefore, AI-assisted DDL and traditional DDL are structurally similar in learning logic even if their interfaces differ.

Traditional DDL positions learners as query initiators who must formulate search criteria, execute corpus queries, and interpret concordance results. This learner-driven approach fosters autonomy but imposes high cognitive demands (Sun and Wang 2003). Moreover, corpus evidence reflects what has been written, not necessarily what could be written, a limitation that constrains creative writing where novelty and surprise are pedagogical goals. Generative AI transforms this dynamic by enabling system-initiated pattern presentation. Rather than learners laboriously searching

corpora, AI proactively generates lines and alternatives that instantiate linguistic patterns within the creative process. This shift from learner-initiated queries to system-initiated presentation reduces cognitive overload while raising new questions: how should AI-mediated pattern presentation be configured to provide appropriate scaffolding for novice writers without displacing authorship?

This study proposes a three-interaction-types framework that integrates DDL principles, scaffolding theory, and human-AI collaboration research. Coenen et al. (2022) identify serendipitous discovery as crucial for productive human-AI collaboration. Qian et al. (2023) show AI can redistribute cognitive load. Li (2025) finds AI shifts writers toward strategic collaboration. However, these studies treat AI as monolithic rather than examining how different AI configurations produce distinct interaction patterns. Our framework reveals the mechanism through which AI can dynamically adapt scaffolding: parameter configuration, specifically temperature and top-p settings, systematically shapes interaction type distributions. Lower parameter values (e.g., temperature 0.3, top-p 0.4) produce deterministic, corpus-typical outputs; higher values (e.g., temperature 0.8, top-p 0.9) generate more exploratory outputs (Holtzman et al. 2019; Li et al. 2025; Peeperkorn et al. 2024). This technical mechanism enables pedagogical adaptivity: parameter manipulation functions as a design lever for shifting between different DDL scaffolding modes.

We theorize three interaction types as parameter-dependent DDL-scaffolding modes, each serving distinct pedagogical functions:

Type A Constraint Repair: AI identifies deviations from linguistic or structural patterns and provides corrective guidance. This parallels traditional DDL use of concordance lines but occurs proactively rather than through learner-initiated queries. Type A represents normative pattern reinforcement, supporting formal requirements like syllable counts or grammatical accuracy. Low parameter values maximize Type A, providing heavy scaffolding but risking constraint on creative autonomy.

Type B Exemplar Giving: AI provides model phrases or line alternatives that learners select and adapt, aligning with imitation-transformation pedagogy (Hanauer 2010). Type B reduces cognitive load by offering concrete starting points and diverse stylistic models. It supports accessibility but risks passive selection over active creation.

Type C Surprise Harvest: AI generates unexpected ideas, metaphors, or directions that learners would not discover through conventional corpus consultation, aligning with serendipitous discovery in human-AI co-creativity (Coenen et al. 2022) and with theories of divergent thinking and the role of novelty in creative processes (Boden 2004). High parameter values increase Type C frequency, enabling AI to function as pattern-extender rather than pattern-reinforcer. Type C is low-directive and high-autonomy, providing light scaffolding that supports creative expansion while demanding critical judgment from writers.

Our research is guided by one central question and two sub-questions:

RQ1: In university creative writing classrooms, how do different parameter configurations affect the distribution of interaction types (Constraint Repair, Exemplar Giving, Surprise Harvest) in AI-assisted poetry writing?

RQ1a: How do these interaction types influence novice writers' authorship and ownership, creative agency, and satisfaction in collaborative poetry writing?

RQ1b: What pedagogical implications emerge for configuring AI as adaptive DDL scaffolding in creative writing classrooms?

2. Methodology

We designed and deployed a poetry writing platform (poetry.aitutor.ink) that transforms the theoretical framework established in our introduction into a living experimental environment.

The web-based platform features a prompt-designed chatbot system with two critical design principles distinguished from commercial AI tools: 1) Commercial AI tools typically position themselves as content generators. Through careful prompt engineering, we constrained the AI to function as a guide rather than ghostwriter. 2) Where commercial AI tools treat parameters as trade secrets, our platform exposes them as manipulable pedagogy. We engineered the system to function as both learning environment and research instrument. In this way, we can trace the causal pathways through which technical configurations translate into learner experiences. The platform was built using Lovable with OpenRouter API integration, enabling access to GPT-4 as the underlying language model while maintaining full control over generation parameters.

We crafted prompts that would govern AI behavior across chatrooms, making it impossible to discern whether observed variations stem from how the AI is configured (parameters) or what it is instructed to do (prompts). By holding the prompts while manipulating parameters, we created a clean experimental contrast where differences in interaction patterns could be attributed to different scaffolding strategies under various parameter conditions.

Our prompt defines the chatbot's role as a poetry writing assistant for learners and establishes response frameworks supporting all three interaction types:

1. Type A (Constraint Repair): Diagnostic feedback on structural, linguistic, or poetic elements requiring attention;
2. Type B (Exemplar Giving): Model texts, phrase alternatives, and curated options for learner selection and adaptation;
3. Type C (Surprise Harvest): Creative suggestions, unexpected metaphors, and generative possibilities extending beyond learner's initial ideas.

The prompt establishes clear functional boundaries, while encouraging diverse engagement. Rather than positioning the AI as a corrector or an instructor, the prompt frames it as a collaborative partner in the learner's creative process.

By establishing two chatroom with different parameter conditions, we re-conceptualizes temperature and top-p as pedagogical design instruments that educators can use to scaffold:

Structured Chatroom: The Steady Guide (low parameter condition)

Temperature: 0.3; Top-p: 0.4

Technical function: Tightens the AI's decision-making scope, producing predictable, corpus-typical outputs.

DDL function: Transforms the AI into a proactive tutor, surfacing linguistic patterns that learners would encounter through traditional corpus searches but delivering them contextually and conversationally.

Scaffolding function: Generates a teaching style dominated by Type A normative corrections and Type B conventional exemplars, creating heavy scaffolding that reduces cognitive load but risks constraining creative autonomy.

Exploratory Chatroom: The Creative Catalyst (high parameter condition)

Temperature: 0.8 | Top-p: 0.9

Technical function: Loosens the AI's generative constraints, sampling from broader token distributions, enabling outputs that deviate creatively from corpus norms.

DDL function: Generates linguistic possibilities that transcend training data frequencies.

Scaffolding function: Shifts the interaction ecology toward Type C exploratory divergence, providing light scaffolding that sparks creative expansion.

This parameter-interaction design reveals the mechanism through which AI can dynamically adapt to pedagogical circumstances, transitioning from structured tutor to exploratory partner simply by adjusting two numerical values. The how of dynamic DDL scaffolding becomes operationally tractable.

This study employed a three-phase design investigating AI-mediated DDL scaffolding across controlled and naturalistic classroom contexts:

Phase 1: 2x2 Factorial Experiment (N=10)

Design: Controlled laboratory-style experiment manipulating two independent variables:

1. Parameter Configuration (between-subjects)

Structured: Temperature 0.3, Top-p 0.4 (Rooms A and B)

Exploratory: Temperature 0.8, Top-p 0.9 (Rooms C and D)

2. Awareness Condition (between-subjects)

Aware: Participants are aware of parameters and potential effects (Rooms A and C)

Unaware: Participants do not know parameters or their effects (Rooms B and D).

This factorial design enables examination of direct parameter effects and meta-cognitive influences of parameter literacy on authorship and satisfaction.

Phase 2: Natural Classroom Environment: Offline In-Person Workshop (N=8)

Design: An offline classroom setting where participants self-select parameter conditions based on personal preference and creative goals. This phase investigates how writers engage with AI-assisted poetry writing when given agency, examining whether Phase 1 findings hold in authentic classroom practice.

Phase 3: Natural Classroom Environment: Online Workshop (N=12)

Design: An online learning environment maintaining self-selected parameter conditions. This phase extends Phase 2 across different learning environments to test robustness in remote interaction. The progression from controlled experiment (Phase 1) to naturalistic classroom contexts (Phases 2 and 3) strengthens both internal validity and ecological validity.

Context and Participants

We recruited 30 university students from creative writing classrooms through course announcements and digital posters. Participants ranged from novice to intermediate writers with varied prior exposure to poetry and AI tools. The study emphasizes creative writing pedagogy rather than language background as a primary variable.

Intervention and Procedure

All three research sessions unfolded as a 75-minute experience designed to immerse participants in collaborative poetry writing:

Phase 1: Platform Orientation (10 minutes) Participants logged into assigned or self-selected rooms, familiarized themselves with the chatbot interface.

Phase 2: AI-Assisted Poetry Writing (35 minutes) Participants drafted, revised, and refined poems in conversation with the AI partner.

Phase 3: In-class Reflection Template (15 minutes) Participants reflected on authorship perception, creative satisfaction, and scaffolding experiences.

Phase 4: Panel Discussion (15 minutes) Participants shared insights on parameter effects and collaborative writing processes.

Data Collection

We collected five sources of data:

1. Complete chat histories from each participant.
2. Interaction type coding that classified each AI response according to the three interaction types.

3. Poem artifacts stored on the platform.
4. Reflection templates on authorship, satisfaction, perceived helpfulness, AI role descriptions, and preferences.
5. Panel discussion transcripts capturing emergent themes and metaphors.

Data Analysis Plan

Quantitative Analysis

RQ: Do parameter configurations generate distinct interaction type profiles?

We compute interaction-type frequencies and percentages within each parameter condition and compare the resulting profiles across structured versus exploratory settings.

Qualitative Analysis

We apply conversation analysis (Jefferson 2004) to examine uptake patterns and scaffolding trajectories, and discourse analysis to examine how participants narrate identity and agency in panel discussions.

Ethical Safeguards and Transparency

All participants provided consent. Chat logs and personal information were stored in separate databases following IRB protocols. AI tool versions, prompts, and usage policies were documented for transparency.

3. Results

Across three phases, 1000+ message exchanges, 29 reflection templates, and 60+ minutes of panel discussion transcripts, we observe three core results:

1. Parameter Configuration Functions as Learner-Intuitive Scaffolding

Writers developed a reliable sense for different AI scaffolding intensities and could identify which parameter conditions aligned with their creative goals. Structured rooms generated interactions dominated by constraint repair (Type A) and exemplar giving (Type B), while exploratory rooms increased surprise harvest (Type C). These distributions were stable across controlled and naturalistic contexts.

2. The Type B Paradox: Helpful but Alienating

Exemplar-giving interactions were consistently rated the most frequent and most helpful, yet also the most likely to trigger feelings of reduced ownership when adopted uncritically. This paradox underscores a tension between efficiency and authorship in AI-assisted writing.

3. Type C Enables Co-Authorship Through Iterative Reciprocity

Creative agency and satisfaction emerged most strongly in Type C (Surprise Harvest) interactions, when writers receive creative ideas from AI and critically engaged with AI suggestions, including rejecting, negotiating, and refining them. These iterative reciprocal moves were most frequent in exploratory settings and were strongly associated with perceived ownership.

Across the controlled experiment and two naturalistic classroom phases, parameter settings reliably shaped interaction type distributions and were experienced by learners as distinct scaffolding modes. In Phase 1, the structured condition ($T=0.3$, $\text{top-p}=0.4$) produced a tutor-like profile dominated by Type A constraint repair (60%) and Type B exemplar giving (35%), while Type C surprise harvest appeared only 5%. In the exploratory condition ($T=0.8$, $\text{top-p}=0.9$), Type C rose to 35%, a 7-fold increase ($\chi^2 = 24.3$, $p < .001$, Cramer's $V = .38$), indicating a robust shift toward divergent, low-directive collaboration (Figure 1).

When participants could choose freely, their behavior echoed these distributions. In Phase 2, 75% voluntarily selected the exploratory room; in Phase 3, 66.7% chose exploratory, 8.3% chose structured, and 62% switched between modes within a single session. These convergent patterns suggest that parameter configuration is not merely a technical setting but a phenomenologically real scaffold that learners can detect, interpret, and strategically deploy through felt interactional texture. As participants

described, exploratory AI offered “random non-related stuff” that sparked new thinking (P12), while structured AI felt “like a menu” of options (P14). Together, the evidence supports the claim that parameter manipulation functions as an intuitive pedagogical lever that regulates which interaction types become available and how authorship and agency are experienced. The shift from structured to exploratory parameters doesn’t merely alter statistical distributions in coded transcripts; it transforms the phenomenological character of collaboration from “patient tutor” to “imaginative collaborator.”

During Phase 2’s panel discussion, participants spontaneously theorized parameter effects through lived experience, developing phenomenological categories that aligned remarkably with our analytical distinctions:

P12 (exploratory condition): “It gives me random non-related stuff so I could think more about it”, hinting Type C’s serendipitous discovery function.

P14 (structured condition): “Multiple options to choose from... very structured, like a menu”, capturing Type B’s exemplar-selection dynamic.

Key insight: Parameters function as learner-intuitive pedagogical levers, consistently determining which interaction types become available, how learners perceive authorship, and what scaffolding feels appropriate for different creative goals.

Type B interactions, the exemplar-giving, consistently emerged as both pedagogically valuable and creatively alienating. Participants voted Type B “most common” (61.54%) and “most helpful” (55.56%) more than any other interaction type (Type A 19.23% Type C 19.23% for most commonly used type and Type C 33.33% Type A 11.11% for most helpful) (see Figure 2), acknowledging its frequency and usefulness in AI-assisted writing.

Why participants found Type B “Helpful”? We categorized the reasons for favoring Type B into five aspects: 1) Confidence booster 2) Convenient tool 3) Certainty creator 4) Vocabulary support 5) Knowledge gap-filler.

Participants/Phases of experiment	Why Type B is the most helpful interaction type in writing	Quotes from feedback templates and Discussion sessions
P 14 from phase two	Type B as confidence booster	“It (type B) helped me build confidence in choosing words”
P 03 from phase one	Type B as a convenient start for poetry writing	“Because it guided me... and made poem writing easier for me”
P 09 from phase one; p18 from phase two	Type B creates certainty	“It (type B) is clear, all that I need to do is just to choose;” “Because it directly gave me certain options which significantly

		helped with idea/sentence generation”
P 15 from phase two	Type B as vocabulary support	“...lexical enhancement and vocabulary support”
P 23 from phase three	Type B fills the gap in language knowledge	“It covers the language corner I am not aware of”

Yet those who experienced Type B reported diminished authorship perception and less satisfaction towards the output, creating a tension between utility, creativity and ownership that persisted across every methodological variation.

Participants describing Type B interaction:

P 02: “I do not feel ownership or pride over much of it”

P 05: “It (type B) feels like just AI... I couldn’t feel any sort of creativity”, suggesting the sense of being reduced from creator to content provider, watching the AI transform feeling into verse without genuine collaboration.

P 06: “ After it (type B), the poem became more formulaic,” noticing the loss of emotional authenticity in the translation from expression to formulaic product.

P 17: “Looking at the vocabulary choices (provided by AI), it seems somewhat superficial compared to work by real artists.”

This is a specific example of Type B interaction, evidenced in our chat transcripts from structured chatroom:

1. AI presents option set: “Here are a few options for your next line: (a) Rabbits scurry with visions in tow; (b) Machines hum softly, stitching joy and woe; (c) Every dream takes shape, in colors bright and bold”
2. Learner selects: “I’ll use option (b)”
3. Pattern incorporated without deep processing: The chosen line gets integrated into the poem without interrogation of why “stitching” creates particular effects, how “joy and woe” functions as binary construction, or what alternatives might exist beyond the three options presented.

Key insight: This sequence of dialogue optimizes for task efficiency. Learners rapidly choose a competent-looking line, but bypasses the transformative struggle that builds creative and linguistic capacity. The learner never contemplate the generative question: How do I create a line that captures my specific meaning with distinctive voice? Instead, they answer the selective question: Which pre-made line fits adequately? The cognitive demand drops from generation to evaluation, from synthesis to selection.

If Type B represents AI as helpful yet alienating collaborator, Type C emerges as something different: AI as a partner in genuine co-creation. Across all three phases, participants who experienced high frequencies of Type C “surprise harvest” claimed substantial authorship (60-96% range) (see Figure 3) compared to those experienced Type B. They reported more creative satisfaction, and described AI as “partner”.

The mechanism proves to be iterative reciprocity: creativity, authorship and satisfaction in writing emerges from AI’s unexpected suggestions and participants’ critical engagement with these suggestions in the process. Here are some evidence:

Phase 1: When Surprise Was Engineered

Chat log analysis revealed P 09 engaged in sustained Type C interactions:

1. Initial conceptual exploration: AI posed philosophical questions about criminal justice reform themes;
2. Craft innovation: AI suggested “found poetry” technique incorporating legal citations;
3. Iterative refinement: AI offered structural feedback while honoring the participant’s creative vision;
4. Authorial control: Participant rejected, modified, and synthesized AI suggestions selectively.

This participant’s authorship rationale (feedback form) shows that: “Yes [authorship matters], otherwise there is no point in the author being involved at all”

Poem Outcome: “Turnstile Justice” is a 13-line experimental piece incorporating legal metaphors, structured stanzas, and emotional interrogation of systemic failures:

Turnstile justice, turnstile court,
Mercy for him, but what of me?
What of us all, sheep among wolves,
A gavel rung, a soothing wave,
But for whom?

An offset sentence, bailed out for naught,
Promises of reform, with rehabilitation,
But the system let them back in, rested,
Bold to this pen of us sheep, who die for this test.

The wolves lunge forth, fresh from the bench,
A culling of those who were never asked,
I weep as I see, turnstile justice,
Whom does it serve?

The poem demonstrates sophisticated craft beyond typical novice work, suggesting Type C interactions can scaffold upward toward advanced competence rather than merely supporting basic production. The Participant Claimed 80% self-authorship.

Phase 2: When the Participants are Critically Engaged

P12, who chose the exploratory chatroom, claimed 90% self-authorship, the highest ownership perception across all 30 participants, despite AI contributing an estimated 15-20% of the final words. How could such high authorship coexist with substantial AI involvement? The chat logs tell the story through five distinct iterative cycles that reveal the essence of Type C collaboration:

1. Initial creative vision: “I want to write about a cursed princess in dragon form”, giving the conceptual idea/theme, entirely human-generated;
2. AI’s first attempt: Opening lines about transformation and curse;
3. Critical evaluation: “I don’t want receptive use of the word whisper,” rejecting AI’s linguistic choice while accepting the structural approach;
4. AI refinement: Adjusted to “hidden hope,” incorporating feedback while maintaining poetic register;
5. Narrative clarification: “The princess seeks the witch’s help, not vice versa,” correcting AI’s misunderstanding of character agency and plot direction.

P12 maintained critically engaged throughout, steering AI toward their creative vision through evaluative feedback rather than simply accepting or rejecting complete outputs. Each cycle represented active guidance, not passive selection.

P16 reported a balanced co-creation of 50% self, 50% AI with maximum satisfaction (5/5). During panel discussion, the participant narrated the creative process: “I asked the AI to write about ‘reality hurts.’ Then I instructed it to start from the stage where a person is defeated... From there, I guided it through the next step: training and revenge, and finally to the last step: success.” The verb choices matter deeply: “asked,” “instructed,” “guided,” language of choreography, not consumption. The participant didn’t passively select from AI-generated options; they choreographed the creative arc, with AI providing linguistic and formal realization of human-specified emotional trajectory.

Phase 3: When Type C Brings Creativity

Full autonomy in Phase 3 revealed individual differences in how learners experienced Type C interactions, demonstrating that the same interaction type can serve multiple creative functions.

P 27 (Type C preference) demonstrated that high satisfaction can coexist with modest authorship, the participant reported authorship to be “30% mine, 70% AI.” yet rating AI partnership and emotional safety at maximum (5/5). The explanation explains why: “I’m not a very creative person, but I feel like I can rely on AI to have some sparks in my brain,” valuing AI precisely for cognitive augmentation, treating the collaboration as distributed creativity where AI contributes generative capacity that the human lacks. High AI involvement didn’t diminish satisfaction because the participant experienced it as collaborative enhancement rather than creative displacement.

4. Discussion

This discussion interprets our cross-phase findings through four lenses: (1) how the Parameter→Interaction→Perception mechanism operationalizes dynamic DDL

scaffolding, (2) why the Type B paradox reveals fundamental tensions between surface and deep learning in AI-mediated creative pedagogy, (3) how Type C interactions enable DDL-compatible collaboration through provocation and negotiation, and (4) what these mechanisms suggest for designing AI tools that preserve learner agency while delivers support. Through this analysis, we answer that, DDL's future might not depend on AI's capabilities but on educators' capacity to recognize and exercise pedagogical judgment over AI configuration.

The seven-fold increase in Type C interactions when moving from structured ($T=0.3/Top-p=0.4$) to exploratory parameters ($T=0.8/Top-p=0.9$) represents more than a statistically significant finding. It demonstrates parameter configuration as a pedagogical lever with measurable effects. When parameters shift from low to high, AI's phenomenological character changes from a tutor, enforcing corpus-typical patterns, to collaborator extending beyond predictable conventions. In structured rooms where low parameters constrained AI to high-probability token selections, learners met an AI that behaved as meticulous editor, catching deviations from conventional patterns, offering polished alternatives drawn from training data frequencies, providing linguistic safety nets that prevented errors but also constrained creative risk-taking. The interaction ecology became dominated by Type A constraint repair (60%) and Type B exemplar selection (35%), creating what participants characterized as "grammar checker" (P 06) and "mechanical" (multiple participants) assistance.

In exploratory rooms where high parameters enabled broader token sampling, learners encountered something qualitatively different: an AI that posed philosophical questions about their creative intentions, suggested unexpected craft techniques like "found poetry" incorporating legal citations, ventured creative risks through surprising metaphors, and responded to emotional vulnerability with nuanced recognition rather than formulaic consolation.

This difference operationalizes what traditional Data-Driven Learning might be hard to achieve. Johns' (1991) DDL vision positioned learners as active "research workers" discovering patterns through concordance interrogation, but engagement intensity remained essentially uniform. Learners encountered whatever patterns their queries surfaced from static corpora. AI-mediated DDL enables system-initiated engagement intensity modulation where parameter settings influence whether learners encounter predominantly corpus-typical patterns (structured parameters prioritizing high-frequency, conventional outputs) or corpus-extending provocations (exploratory parameters sampling from broader, more diverse token distributions).

To understand why Type B helpfulness and alienation coexist, we must examine the cognitive processes Type B interactions elicit through the lens of surface versus deep learning (Ramsden 2003). This distinction illuminates what Type B optimizes and what it undermines: When AI presents predetermined exemplars that learners can select and use without interrogating how the patterns work or why certain choices matter, the cognitive process becomes reproductive selection rather than generative construction (Montag & Elhai, 2025). Learners engage in pattern-application (choosing appropriate exemplars from available options) rather than pattern-discovery

(understanding why patterns work and how to create novel instances). In surface-oriented uptake, the learner's task collapses into choosing among options rather than interrogating why a line fits the poem's intent, voice, or imagery.

Assessment implications follow directly from this distinction. If surface uptake is possible through Type B, then evaluating AI-assisted writing solely by product quality risks rewarding compliance rather than depth of engagement. Assessment should therefore attend to process evidence: rationale for selections, revisions made after AI suggestions, and reflections that demonstrate evaluative control. Rubrics can explicitly credit the learner's critical decision-making and the degree to which AI output was interrogated, reshaped, or rejected. Such criteria make deep engagement visible and reduce incentives for passive adoption. This also creates alignment with DDL values, ensuring that AI functions as data for analysis rather than as a shortcut to completion.

If Type B represents the downside of AI scaffolding, type C interactions might prove its transformative potential. Across all three phases, participants who experienced high frequencies of Type C "surprise harvest" interactions demonstrated a higher sense of authorship and satisfaction architecture.

The process is evident throughout our experiment: When AI provides unexpected metaphors and ideas requiring judgments, the reciprocal cycle of rejection-refinement becomes an assertion of creative agency. The learner exercises critical engagement at every decision point, they feel emotionally more in control of the learning process, thus the high satisfaction rate. While Type B requires only selection (Which pre-made option that appeals more adequately?), Type C requires synthesis (How do I integrate this AI suggestion into my poem?), which might in a certain way transfer surface learning to deep learning.

We frame Type C's synthesis using a creativity framework that integrates Guilford's (1950) divergent and convergent creativity, Boden's (2003) theory of combinational creativity, and empirical evidence on serendipitous discovery (Coenen et al. 2022). Divergent thinking generates multiple possibilities, while convergent thinking evaluates and refines them. Type C interactions supply divergent possibilities by extending the option space beyond what a learner in writing would generate alone. Boden argues that computational systems can expand conceptual spaces by making non-obvious combinations; Type C operationalizes this by introducing surprising metaphors and directions. Empirically, Coenen et al. show that serendipitous discovery supports productive human-AI collaboration, a pattern observed here as writers used unexpected AI outputs to spark new lines, contrasts, and imagery. This tension transforms AI from pattern-retriever to pattern-extender. Traditional corpus DDL operates within a corpus-bounded possibility space, discovering patterns that exist in training data, frequencies that reflect what has been written. This is important for developing corpus-typical competence (knowing what educated native writers conventionally do), but it constrains creative exploration to reproducing existing patterns. Type C, enabled through exploratory parameters that sample from broader token distributions, extends beyond corpus frequencies into a generative possibility space. The "random non-related stuff" that P12 valued represents AI's capacity to

suggest connections, techniques, and framings emerging through novel pattern recombination rather than corpus retrieval. These provocations are unlikely to appear in traditional concordance searches because they do not exist as high-frequency patterns, emerging through creative divergence.

However, novelty alone does not guarantee agency. The data show that agency grows when writers move from receiving surprise to critically engaging with it: evaluating relevance, revising language, and integrating or rejecting suggestions. This critical engagement builds self-efficacy because writers experience themselves as decision makers rather than passive acceptors. Previous research has demonstrated similar result. McQuire et al. (2024) show that engagement declines when learners receive an AI-generated poem and are asked to edit it, yet creativity grows back when they co-create with AI and establish self-efficacy. The iterative cycle of generate, judge, revise, and refine becomes a DDL-like loop that reinforces ownership. In other words, Type C contributes to agency not because it replaces human choice but because it provokes reflective choice.

This pedagogical framing aligns with UNESCO's AI Competency Framework for Education (2024), which emphasizes a human-centred mindset: AI must strengthen human capacities, and selection, use, and monitoring of impact must remain human-controlled and accountable. In AI-assisted poetry writing, the human-centred outcome is what learners gain: deeper understanding of poetics, wider expressive techniques, more fluent use of AI as a tool for articulating artistic experience, and stronger ability to discern and evaluate co-created output. For educators, the task is to design the classroom ecology so that AI functions more as a Type C role, supporting, prompting, inspiring, while self-efficacy and human agency remain the centre of the creative process.

Our findings provide what we might call a conditional resolution to the challenge about whether generative AI represents DDL's death or evolution. The answer is neither predetermined nor technologically determined. It depends on how educators understand, configure, prompt, and design pedagogical AI systems.

Under structured parameters that constrain AI to corpus-typical outputs, AI might be helpful for creative writing learners, but functions as DDL's antithesis. It generates predetermined exemplars that position learners as selectors rather than discoverers, promoting surface learning through reproductive selection rather than deep learning through generative construction, enabling task completion while bypassing the transformative struggle that builds creative and linguistic capacity.

Under exploratory parameters that enable corpus-extending provocations, AI functions as a potential evolution, generating unexpected possibilities that position learners as co-constructors exercising selective uptake and critical judgment, promoting deep learning through synthesis and integration rather than surface learning through selection, scaffolding creative expansion while demanding continuous directional control. AI extends the work available to learners, expanding the possibility space beyond what corpus consultation could provide while preserving the agency and active construction central to DDL principles.

Parameter settings are not technical specifications but pedagogical design variables determining whether AI constrains or enables, scaffolds appropriately or over-determines outcomes, promotes surface learning or supports deep engagement. This insight transforms the question of responsible AI integration from whether to use AI (presence is increasingly inevitable as these tools proliferate) to how to configure AI serving learning goals rather than undermining them.

Design Recommendations for Parameter-Aware Pedagogical AI

Our findings establish parameter literacy as essential professional competence for educators integrating AI into writing instruction. This literacy encompasses three interconnected dimensions:

Technical Understanding: Educators need functional knowledge of how temperature and top-p settings shape AI behavior, understanding that lower values produce deterministic, corpus-typical outputs while higher values enable variable, exploratory directions. This knowledge should become standard content in educational technology courses and professional development programs.

Pedagogical Judgment: Educators must exercise judgment about when structured versus exploratory parameters serve learning goals, recognizing that different creative tasks, proficiency levels, and developmental stages may require different scaffolding intensities. This judgment develops through experience experimenting with configurations and observing how learners respond.

Tool Selection: Educators need capacity to identify AI tools that expose parameters for educator control versus those that hide parameters as proprietary settings. This requires advocating for transparency in educational AI design and potentially rejecting tools that don't provide necessary configurability.

Our findings suggest progressive parameter curriculum implementing scaffolding fade through systematic configuration adjustment:

Weeks 1-3 (Foundation Building): Structured parameters ($T=0.3/Top-p=0.4$) provide heavy scaffolding as learners develop basic competence with poetic forms, master fundamental techniques, and build confidence with AI interaction. Type A constraint repair dominates, establishing technical foundations.

Weeks 4-6 (Voice Development): Balance Type B exemplars with emerging Type C provocations as learners begin developing personal style while maintaining support. Scaffolding intensity decreases gradually.

Weeks 7-9 (Creative Independence): Exploratory parameters ($T=0.8/Top-p=0.9$) enable Type C-dominant interactions as learners take creative risks with minimal constraint. AI functions as provocative collaborator rather than directive tutor.

Limitations

Our three-phase study, while providing cross-context evidence, carries limitations requiring acknowledgment. The sample comprised 30 learners within a specific institutional context, limiting generalizability across diverse populations, proficiency levels, cultural contexts, and educational systems. Our participant pool consisted entirely of L2 learners, so we have not yet examined potential differences between L1 and L2 writers in how parameter-based scaffolding is perceived or used. Future work should extend to longer-term classroom integration, explore instructor interventions that prompt critical engagement with AI output, and refine measures of authorship and self-efficacy. Further research can also examine how parameter shifts interact with genre, task type, and peer collaboration.

5. Conclusion

Generative AI represents neither the death of Data-Driven Learning nor its simple continuation, but a pedagogical revolution where parameter configuration determines which path unfolds.

With structured parameters constraining AI to corpus-typical patterns and Type B exemplar selection, AI creates “helpful but alienating” experiences where task completion occurs at authorship’s expense, where surface learning replaces deep engagement. With exploratory parameters enabling corpus-extending provocations and Type C surprise harvest, AI enables DDL evolution, preserving and amplifying creative agency while providing transformative support through provocations that extend possibility space beyond what concordance searches could surface. Learners exercise directional control through iterative reciprocity, develop procedural ownership through rejection-refinement cycles, and maintain authentic voice through strategic selective uptake of AI suggestions. The “hands-on” engagement central to DDL pedagogy persists, transformed from querying static corpora to evaluating dynamic provocations.

The pedagogical imperative becomes clear: recognize parameter configuration as pedagogical decision-making, make it visible and adjustable, design for learner autonomy and metacognitive awareness, and prioritize low-stakes voluntary contexts enabling authentic expression. DDL’s future depends not on AI’s capabilities but on educators’ capacity to exercise informed pedagogical judgment over its configuration. The evolution we demonstrate becomes possible when educators recognize AI parameters as pedagogical levers and design accordingly, making it visible, adjustable, and pedagogically grounded transforms AI from potential threat into powerful partner in DDL’s ongoing evolution in the generative era.

References

- [1] Crosthwaite, P., Baisa, V.: Generative AI and the end of corpus-assisted data-driven learning? Not so fast! *Applied Corpus Linguistics* 3(3), 115 (2023)
- [2] Johns, T.: Should you be persuaded: Two examples of data-driven learning. *Classroom Concordancing Newsletter* 4, 1–16 (1991)
- [3] Boulton, A.: Data-driven learning: Taking the computer out of the equation. *Language Learning* 62(2), 534–572 (2012)
- [4] Thorne, S.L., and Lantolf, J.P.: “A linguistics of communicative activity.” *Bilingual Education and Bilingualism* 62, 170 (2006)
- [5] Sinclair, J.: *Corpus, Concordance, Collocation*. Oxford: Oxford University Press, 1991.
- [6] Cobb, T.: “Frequency 2.0: Incorporating homoforms and multiword units in pedagogical frequency lists.” *L2 vocabulary acquisition, knowledge and use: New perspectives on assessment and corpus analysis*, 79-108 (2013)
- [7] Vaswani, A. et al.: “Attention is all you need.” *Advances in neural information processing systems* 30 (2017)
- [8] Radford, A. et al.: “Language models are unsupervised multitask learners.” *OpenAI blog* 1, no. 8, 9 (2019)
- [9] Heidegger, Martin. 1984. Holderlins Hymne “Der Ister”. Gesamtausgabe 53. Frankfurt am Main: Klostermann.
- [10] Strehovec, Janez. “From Poetry As the Excess of Language to Poems Generated by Artificial Intelligence (Poet As Researcher in the World of New Cultural Paradigms).” *Canadian Journal of Language and Literature Studies* 3, no. 4 (2023): 98-111.
- [11] Sun, Y.C., Wang, L.Y.: Concordancers in the EFL classroom: Cognitive approaches and collocation difficulty. *Computer Assisted Language Learning* 16(1), 83–94 (2003)
- [12] Bernardini, S.: Corpora in the classroom: An overview and some reflections on future development. In: Sinclair, J.M. (ed.) *How to Use Corpora in Language Teaching*, pp. 15–36. John Benjamins (2004)

- [13] Hanauer, D.I.: Poetry as research: Exploring second language poetry writing. John Benjamins Publishing Company (2010)
- [14] Lao, C. Y., and Krashen S.: “The impact of popular literature study on literacy development in EFL: More evidence for the power of reading.” *System* 28, no. 2, 261-270(2000)
- [15] Chakrabarty, T., Padmakumar, V., He, H.: Help me write a poem: Instruction tuning as a vehicle for collaborative poetry writing. arXiv:2210.13669 (2022) <https://doi.org/10.48550/arXiv.2210.13669>
- [16] Guilford, J. P. 1950. “Creativity.” *American Psychologist* 5 (9): 444–454.
- [17] Boden, M.A.: The creative mind: Myths and mechanisms. Routledge (2004)
- [18] Coenen, A. et al.: Wordcraft: Story writing with large language models. In: IUI '22 Companion: 27th International Conference on Intelligent User Interfaces Companion, pp. 841–843. Association for Computing Machinery (2022) <https://doi.org/10.1145/3490100.3516505>
- [19] Holtzman, A., Buys, J., Du, L., Forbes, M., Choi, Y.: The curious case of neural text degeneration. arXiv:1904.09751 (2019) <https://doi.org/10.48550/arXiv.1904.09751>
- [20] Li, L. et al.: Exploring the impact of temperature on large language models: Hot or cold? *Procedia Computer Science* 264, 242–251 (2025) <https://doi.org/10.1016/j.procs.2024.10.130>
- [21] Peeperkorn, M., Bewersdorff, A., Zhai, X.: Is temperature the creativity parameter of large language models? arXiv:2405.00492 (2024) <https://doi.org/10.48550/arXiv.2405.00492>
- [22] Qian, W. et al.: “It felt like having a second mind”: Investigating human-AI co-creativity in prewriting with large language models. arXiv:2307.10811 (2023) <https://doi.org/10.48550/arXiv.2307.10811>
- [23] Jefferson, G.: Glossary of transcript symbols with an introduction. In: Lerner, G.H. (ed.) *Conversation Analysis: Studies from the First Generation*, pp. 13–31. John Benjamins (2004)
- [24] Ramsden, Paul. 2003. *Learning to Teach in Higher Education*. 2nd ed. London: RoutledgeFalmer.

[25] Montag, C., and Elhai, J.D.: “Associations between deep learning vs. surface learning styles with both AI competence and personal enhancement by AI.” *Learning and Motivation* 92,102182 (2025)

[26] McGuire, J., Cremer, D.D., and Van de Cruys, T.: “Establishing the importance of co-creation and self-efficacy in creative collaboration with artificial intelligence.” *Scientific Reports* 14, no. 1, 18525 (2024)